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Unit 3 - Designing Solutions

PDF Documents

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UNIT 3 DESIGNING SOLUTIONS

Programming Concepts

WHAT IS A VARIABLE?

Imagine a box. You can put different things inside this box, like a ball, a book, or a toy. In coding, a variable is like a box, but instead of physical objects, we store numbers or words inside. We give each box a name so we can easily find and change what's inside later.

Example

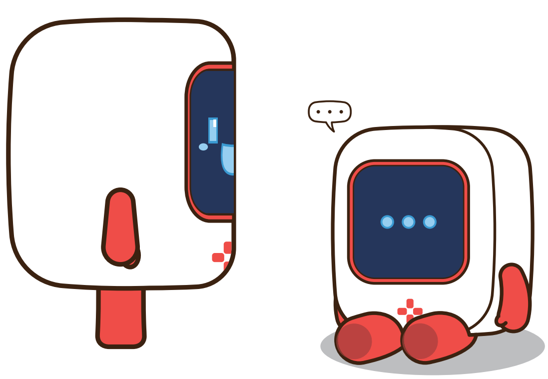
Let's say we want to store a person's age. We can create a variable called "age" and store the number 12 inside it.

WHY DO WE USE VARIABLES?

To store information: We can use variables to store data that we need to use later in our program.

To make our code more flexible, If we want to change a value, we only need to change it in one place (the variable).

To make our code easier to read: Giving variables meaningful names helps us understand what the code is doing.



What is random? When something is random, it happens by chance and is unpredictable. Think of flipping a coin. You don't know if it will land on heads or tails.

Random in coding: We can use a random function to generate random numbers. This is useful for creating games, simulations, or situations where we want something to happen by chance.

Example: We want to create a program that picks a random number between 1 and 10. We would use a random block.

WHY DO WE USE RANDOM?

To create games: Random numbers can be used to make games more interesting and unpredictable. For example, a dice roll or a card shuffle.

To simulate real-world events: We can use random numbers to simulate events like weather or traffic.

To generate random data: Random numbers can be used to generate random data for testing or analysis.

Key points to remember:

A **variable** is a named container for storing values.

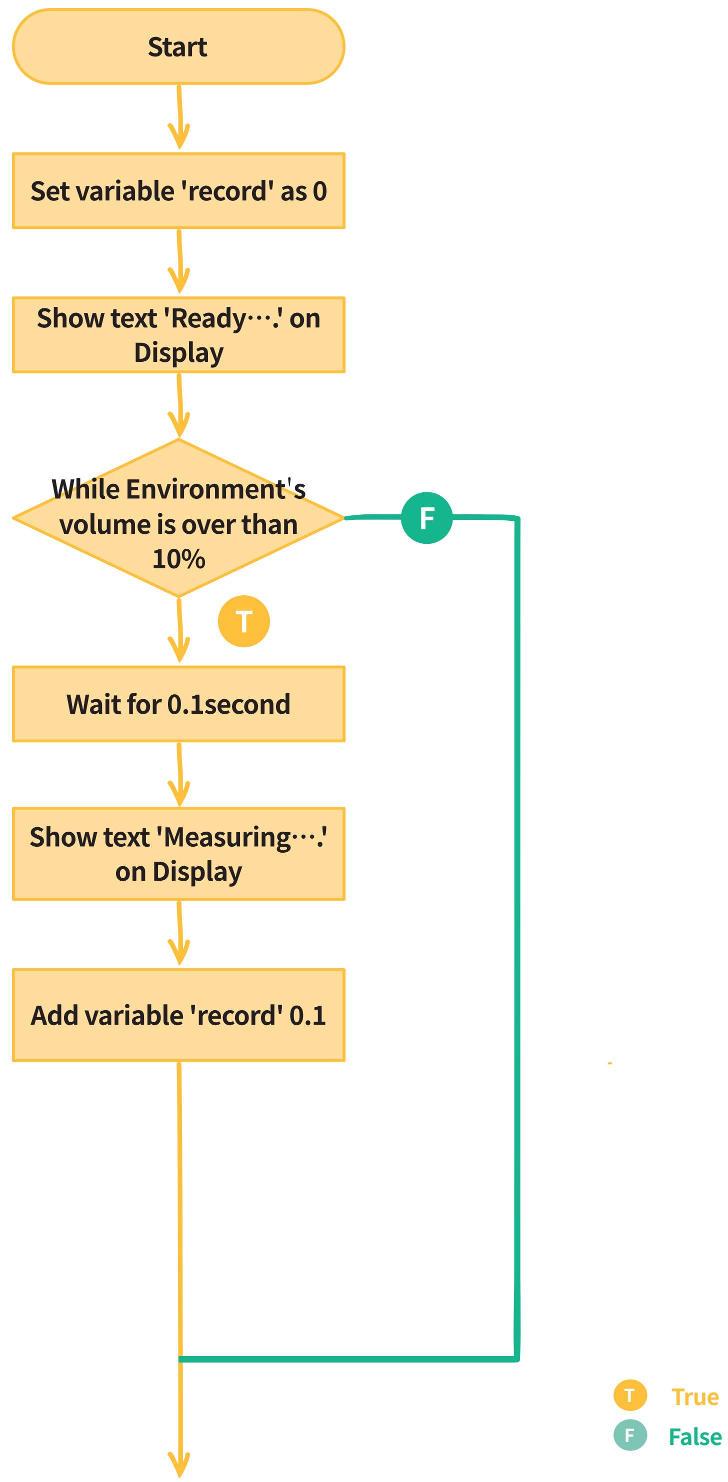
Random means something happens by chance.

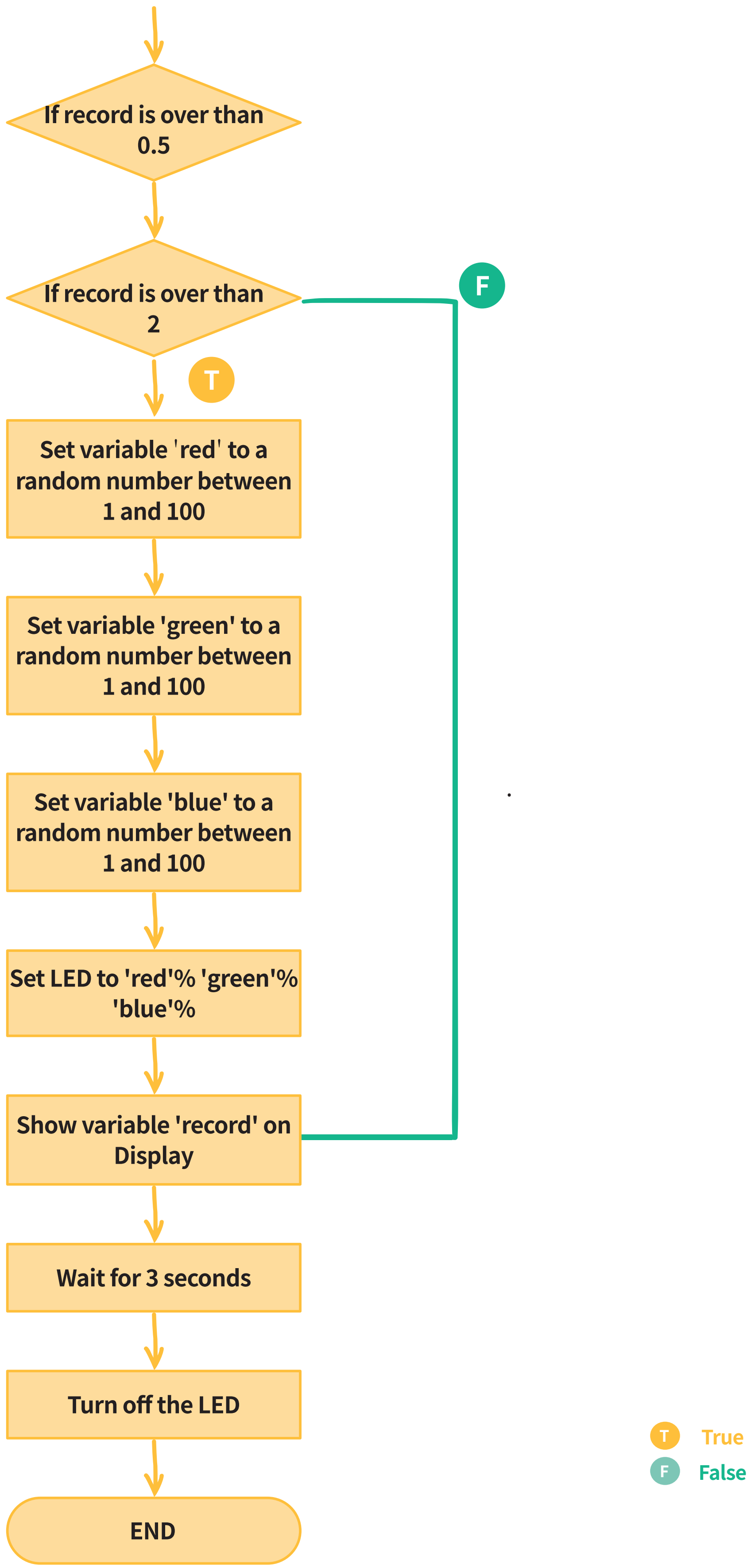
Random functions generate unpredictable numbers.

Flowcharts & Algorithms

Let's read the flowchart and create the code.

FLOWCHART




If record is over than 0.5
If record is over than 2
F
T
Set variable 'red' to a random number between 1 and 100
Set variable 'green' to a random number between 1 and 100
Set variable 'blue' to a random number between 1 and 100
Set LED to 'red'% 'green'% 'blue'%
Show variable 'record' on Display
Wait for 3 seconds
Turn off the LED
END
T True
F False